

MUHAMMAD SOHAIB ASHRAF

Software Engineer Robotics & AI

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PROFESSIONAL SUMMARY

Software Engineer with 5+ years of experience developing production robotics, autonomous systems, computer vision and software-hardware integration. Strong hands-on background in C#, C++, Python, ROS/ROS2, Unity and Linux, with experience spanning robot-control applications, LiDAR/SLAM, point-cloud visualisation, machine learning and deployed robotic systems. Experienced in Git-based development, code review, automated testing, debugging, Linux services and cross-functional engineering. MSc in Artificial Intelligence with Robotics (Distinction) and BSc in Software Engineering.

CORE SKILLS

Programming: C#, C++, Python, SQL, Bash/Shell

Robotics & Autonomous Systems: ROS, ROS2, LiDAR, SLAM, localisation, mapping, navigation, PointCloud2, RViz, RTAB-Map, behaviour trees, robot/device integration

AI & Computer Vision: TensorFlow, deep neural networks, CNNs, semantic segmentation, YOLO, OpenCV, NVIDIA Isaac SDK, visual odometry

Software Engineering: Object-oriented design, SOLID principles, REST APIs, FastAPI, asynchronous/event-driven software, code review, unit testing, integration testing, debugging

Systems & Dev Tools: Linux/Ubuntu, Docker, systemd, Git/GitHub, CMake, colcon, JSON/YAML, configuration, deployment, diagnostics, troubleshooting

Industrial & Hardware Integration: Unity 3D, PLC programming, cameras, sensors, actuators, hardware/software integration, simulation, data analysis and visualisation

PROFESSIONAL EXPERIENCE

Robotics | Design Engineer (Software) | Nexxis | Perth, WA | Jul 2024 - Present

- Develop and maintain production software for industrial robotic inspection systems using C#, Unity, ROS2, Python and Linux, with software deployed on physical robot hardware.
- Deliver cross-repository features across ROS2 interfaces, backend services and operator-facing control applications, integrating robot, sensor and system state across the software stack.
- Work with Argus 3D LiDAR point clouds, SLAM and ROS2 sensor data, including Unity-based 3D visualisation and mapping workflows for robotic inspection.
- Build backend and field-facing services using Python, FastAPI, ROS launch, Docker and systemd; implement configuration-driven robot/device profiles and controls for cameras, actuators and sensors.
- Coordinate guided-operation and robot-configuration changes across application, behaviour-tree and runtime layers, managing interface and compatibility dependencies.
- Improve reliability through automated testing, metadata validation, peer code review and runtime/integration debugging of cross-system configuration, versioning and device faults.

System Engineer | Cube Discipline | Islamabad, Pakistan | Mar 2022 - Oct 2023

- Developed computer-vision and control software for a physical-board AI Go system, detecting game pieces and coordinating AI-selected moves through a magnetic actuation mechanism beneath the board.
- Built machine-vision and automation solutions using C/C++, C# and Python
- Used data analysis and visualisation to evaluate system behaviour and provided technical direction to another engineer within a multidisciplinary team.

Autonomous Systems Engineer | Epazz | Lahore, Pakistan | Mar 2021 - Mar 2022

- Led development of robotics and IoT software using Python, C++, machine vision, machine learning, PLC programming and simulation.
- Trained a TensorFlow deep neural network from scratch on image data to detect and analyse chef movement over time across selected recipes.
- Owned software problem definition and implementation, iterated model and vision behaviour in simulation, and coordinated delivery with software, mechanical and electrical/electronic engineers.

Systems Engineer (Autonomous) | Oneberry Technologies | Singapore | Dec 2019 - Nov 2020

- Developed and deployed computer-vision, localisation and navigation software on autonomous mobile robots using ROS, NVIDIA Isaac SDK, Python and C++.
- Integrated and modified 3D LiDAR and visual-odometry components using ROS PointCloud2, RViz, RTAB-Map and Velodyne tooling, with depth-camera sensing for robotic perception.
- Trained and refined Isaac SDK semantic-segmentation models for sidewalk navigation and implemented turn-speed control to improve stability during sharp manoeuvres.
- Supported integration, deployment and troubleshooting of physical autonomous robotic systems under senior engineering guidance.

Web Developer | Epullon | Lahore, Pakistan | Mar 2017 - Nov 2017

Developed and deployed web-based business applications using PHP, CodeIgniter, JavaScript, jQuery, HTML and CSS, working across backend and frontend development.

Web Developer Intern | Zepto Systems | Islamabad, Pakistan | Jul 2015 - Sep 2015

Worked as a full-stack development intern with emphasis on backend development using PHP, CodeIgniter, JavaScript and jQuery.

PROJECTS

Argus 3D Point-Cloud & SLAM Integration | Nexxis | 2025 - Present

Worked with LiDAR/ROS2 point-cloud data and Unity visualisation; improved mapping/SLAM reliability through node-state handling, timeout/failure states, diagnostics and export workflows.

ROS2 Cross-Stack Robot Integration | Nexxis | 2024 - Present

Delivered software changes across ROS2 interfaces, backend services and operator applications, integrating robot/device state, controls, engineering-unit metadata and configuration across multiple repositories.

IntelliSense - Smart Environment | Personal Project | 2024 - Present

Developing an IoT smart-environment system using Python, machine learning and computer vision to detect emotional cues and trigger context-aware environmental adjustments designed to support user comfort and wellbeing.

Chef Motion Deep Learning | Epazz | 2021 - 2022

Trained a TensorFlow deep neural network from scratch on image data to detect and analyse chef movement over time across selected recipes, iterating the model in simulation.

KenOBI 3D LiDAR & Visual Odometry | Oneberry | 2019 - 2020

Integrated 3D LiDAR, ROS PointCloud2, RTAB-Map, visual odometry and depth-camera sensing for autonomous robot localisation, perception and navigation.

Sidewalk Semantic Segmentation | Oneberry | 2020

Trained and refined an NVIDIA Isaac SDK semantic-segmentation model for traversable-path detection and deployed it on an autonomous mobile robot.

Evolutionary Flocking Bots | University of Hertfordshire | Feb 2019 – May 2019

Developed a simulation-based robotic environment in Unity3D using a Genetic Algorithm to evolve autonomous bot behaviour. The agents learned to locate a target, avoid hazardous areas and improve their strategy across successive generations. Achieved **92%** for the project.

EDUCATION

MSc Artificial Intelligence with Robotics | University of Hertfordshire, UK | 2018 - 2019

Distinction. Focus: machine learning, neural networks, artificial intelligence and robotics.

BS Software Engineering | University of Management and Technology, Pakistan | 2013 - 2017

CERTIFICATIONS

- Software Architecture Foundations | LinkedIn Learning | 2024
- Unity Certified Developer | Unity | 2018